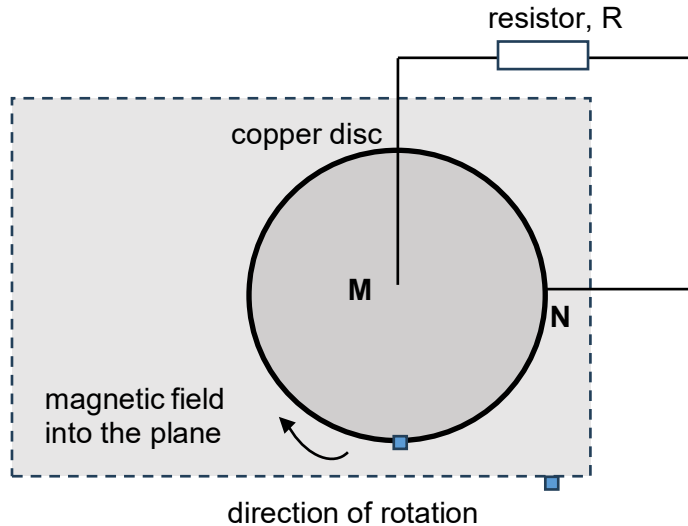


- 24 A copper disc rotates with a constant angular velocity in a uniform magnetic field applied perpendicular to its surface, as shown. The axle and the edge of the disc are in contact with metallic brushes M and N which are connected to a resistor R via wires.



Which of the following statements is correct?

- A There is no current through R because there is no flux change through the disc.
- B There is an alternating current through R.
- C There is a steady current from M, through R, to N.
- D There is a steady current from N, through R, to M.

Ans: D

Draw a line from M to N. When the disc is rotating in a clockwise direction, this line is moving downward. By Lenz's Law, an upward force acts on this line, and by FLHR, the induced current is in a direction from M to N. Hence this current goes from N, through R, to M.

- 25 A 240 V supply is connected to an ideal transformer. The output of the transformer is connected